Exp No:10

Applying Performance Optimization Techniques in Zoho Creator

Aim

To apply performance optimization techniques such as indexing, subforms, and query tuning in a Zoho Creator application to improve data retrieval, processing speed and overall application performance.

Record Status

Completed after applying and testing the performance optimization techniques in the application.

Concept

Performance optimization improves the speed, efficiency and responsiveness of an application.

- Indexing: Helps retrieve records efficiently when supported by the data source and field configuration.
- Subforms: Allow multiple related records to be maintained within a parent record, reducing unnecessary navigation between forms.
- Query tuning: Improves the efficiency of data retrieval by limiting unnecessary searches, filtering records appropriately, and avoiding repeated operations.

These techniques help improve the performance and usability of Zoho Creator applications.

Procedure

1. Open the Loan Application Management System in Zoho Creator.
2. Examine the existing forms, reports, workflows, and Deluge scripts.
3. Identify fields frequently used for searching and filtering loan applications.
4. Review the available indexing options for the application's data source and apply indexing where supported.
5. Explore subforms for related information, such as loan document details or repayment entries, if required by the application.
6. Review Deluge scripts and remove unnecessary repeated data retrieval operations.
7. Apply appropriate criteria to retrieve only the required records.
8. Test the application using sample records before and after optimization.
9. Compare response times and verify that the results remain accurate.

Application Design

The Loan Application Management System can use the following optimization techniques:

- Indexing: Optimize frequently searched fields where indexing is supported.
- Subforms: Group related information, such as document details, under a single loan application.

- Query tuning: Use specific search criteria to retrieve relevant loan applications instead of processing all records.
- Workflow optimization: Avoid unnecessary repeated Deluge operations and redundant updates.
- Report optimization: Display only the fields and records required for the report.

Observations

1. Efficient data retrieval reduces unnecessary processing.
2. Appropriate indexing can improve searches on supported data sources.
3. Subforms organize related records and improve the user experience.
4. Optimized queries reduce unnecessary record retrieval.
5. Removing redundant operations can improve workflow efficiency.

Deluge Script:

```
pendingLoans = Loan_Application[Status == "Pending"];


for each loan in pendingLoans
{
   info loan.Applicant_Name;
   info loan.Status;
}
```

Validation and Discussion

Test the application with sample loan records and verify that searches, reports, and workflows return the correct results.

Compare the response time before and after optimization using the same test conditions. Confirm that subform records are saved correctly and that query changes do not omit required records.

Indexing options depend on the data source and Zoho Creator configuration. Subforms primarily improve data organization and do not automatically guarantee faster performance. Actual performance improvements must be measured.

Result

Thus, performance optimization techniques, including indexing, subforms, and query tuning, were studied and applied where supported in the Zoho Creator application to improve data retrieval efficiency and overall application performance.